



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI KANNIMARA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

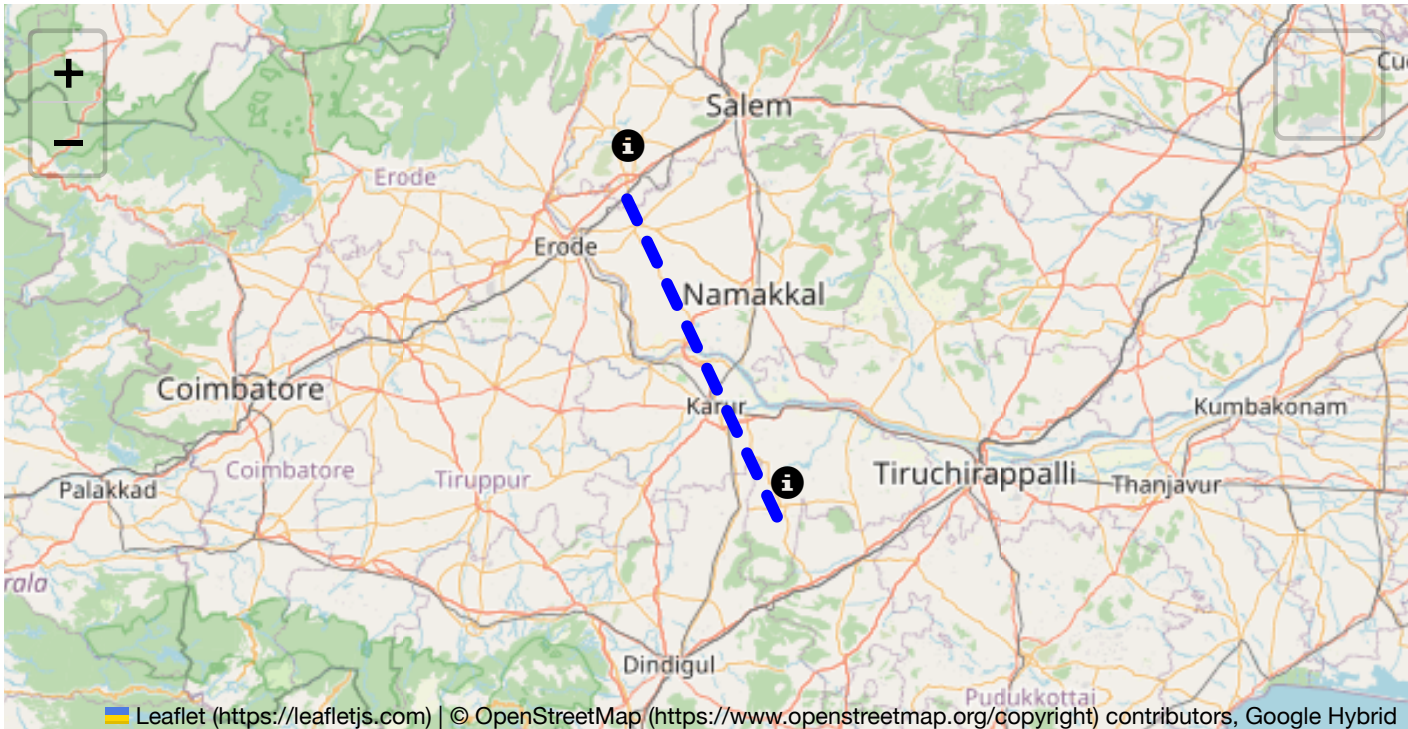
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 113.14 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours

Start: (11.4381, 77.8734)
End: (10.6619, 78.2441)



Welcome to the Journey Risk Management Study

Here's a comprehensive analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu to M66V+XVG, Kurumpapatty, Tamil Nadu, with a focus on safety considerations for a truck carrying hazardous materials:

- Overview of the Route Map:**
 - Starting at Sangagiri, the primary route heads eastward towards Kurumpapatty, crossing through several key towns and highways. Major segments include NH 544 and local roads connecting towns and rural areas. The route is approximately 113.14 kilometers long.
- Typical Weather Conditions and Potential Hazards:**
 - The region generally experiences tropical weather. Summers can be hot, and monsoon rains create potential for flooding and reduced visibility. Between October and December, the northeast monsoon can lead to heavy rainfall, increasing the risk of waterlogging and landslides, particularly in hilly sections.
- Traffic Patterns:**
 - Peak traffic hours are typically around 8-10 AM and 5-7 PM. Congestion is common near urban areas and major junctions, particularly in Salem and the outskirts of Namakkal. Expect delays during these times, especially when navigating through smaller roads.
- Assessment of Road Quality and Infrastructure:**
 - NH 544 is generally well-maintained with good signage and road surfaces. However, secondary roads may have potholes or narrow lanes, posing a challenge for large vehicles. Rural roads closer to Kurumpapatty might lack proper lighting and markings.
- Alternative Routes for Emergencies:**
 - For emergencies, routes via NH 44 can be considered as alternative pathways between major towns. Local knowledge and GPS updates are vital for selecting viable detours, especially during seasonal roadblocks.

6. Local Regulations on Hazardous Material Transport:
- Transport of hazardous materials requires adherence to specific national and state regulations, including proper permits and signage. Restrictions could exist near densely populated areas and environmentally sensitive zones.
7. Historical Incidents Involving Heavy Vehicles:
- Past incidents along this corridor mostly involve vehicular collisions, often due to improper lane discipline and speeding. Heavy vehicles have a slight risk of turnover in certain sharp bends.
8. Environmental Considerations and Sensitive Areas:
- Areas near forest reserves and water bodies require caution to prevent environmental contamination. Special precautions should be in place to avoid leaks or spills in these regions.
9. Communication Coverage:
- Mobile coverage is usually stable along main highways and towns but can be sporadic in rural areas, particularly in hilly terrains. It is advisable to familiarize yourself with areas where communication is unreliable.
10. Estimated Emergency Response Times:
- Emergency services in major towns can respond within 15-25 minutes. However, response times could extend to 45 minutes to an hour in remote parts due to distance and road conditions.
11. Overall Summary of Risk Assessment:
- This route presents moderate risk. Key challenges include occasional extreme weather, traffic congestion, and varying road conditions. Strict adherence to traffic laws, careful planning for potential weather impacts, and preparedness with communication devices are crucial. Awareness of local regulations and environmental guidelines will further ensure the safe transit of hazardous materials.

Drivers and transport coordinators should remain vigilant, employ real-time navigation tools, and maintain regular communication with dispatch centers to handle any emerging issues effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	U-Turn	High	10.9110477, 78.09713769999999	10 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	Medium	11.37902, 77.89500	30 KM/Hr
7	Turn	Medium	11.37850, 77.89453	30 KM/Hr
8	Turn	Medium	11.16755, 78.01568	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14485, 78.02282	15 KM/Hr
11	Turn	Medium	11.10084, 78.00555	30 KM/Hr
12	Turn	Medium	11.08280, 78.00339	30 KM/Hr
13	Turn	Medium	11.05250, 78.05164	30 KM/Hr
14	Blind Spot	Blind Spot	10.90793, 78.09916	10 KM/Hr
15	Blind Spot	Blind Spot	10.91105, 78.09714	10 KM/Hr
16	Turn	High	10.84613, 78.12000	15 KM/Hr
17	Turn	Medium	10.84321, 78.16436	30 KM/Hr
18	Turn	High	10.84057, 78.16518	15 KM/Hr
19	Turn	High	10.83953, 78.18108	15 KM/Hr
20	Turn	Medium	10.83523, 78.18173	30 KM/Hr
21	Turn	Medium	10.82892, 78.20166	30 KM/Hr
22	Turn	Medium	10.82865, 78.20188	30 KM/Hr
23	Turn	Medium	10.79274, 78.20214	30 KM/Hr
24	Turn	Medium	10.79217, 78.20276	30 KM/Hr
25	Turn	Medium	10.79013, 78.20307	30 KM/Hr
26	Turn	Medium	10.78936, 78.20390	30 KM/Hr
27	Turn	Medium	10.78767, 78.20438	30 KM/Hr
28	Turn	Medium	10.78380, 78.20278	30 KM/Hr
29	Turn	High	10.77911, 78.21461	15 KM/Hr
30	Turn	Medium	10.77530, 78.21486	30 KM/Hr
31	Turn	Medium	10.77330, 78.21431	30 KM/Hr
32	Turn	High	10.77239, 78.21467	15 KM/Hr
33	Turn	High	10.77215, 78.21560	15 KM/Hr
34	Turn	High	10.76401, 78.21632	15 KM/Hr
35	Turn	Medium	10.75296, 78.22112	30 KM/Hr
36	Turn	Medium	10.75178, 78.22246	30 KM/Hr
37	Turn	Medium	10.74551, 78.22709	30 KM/Hr
38	Turn	Medium	10.73084, 78.23146	30 KM/Hr
39	Turn	High	10.72910, 78.23136	15 KM/Hr
40	Turn	High	10.72888, 78.23084	15 KM/Hr
41	Turn	Medium	10.71906, 78.23543	30 KM/Hr
42	Turn	High	10.71762, 78.23799	15 KM/Hr
43	Turn	Medium	10.71635, 78.23810	30 KM/Hr
44	Turn	High	10.71463, 78.23992	15 KM/Hr
45	Turn	Medium	10.71217, 78.23933	30 KM/Hr
46	Turn	Medium	10.70648, 78.24053	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
10	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
19	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.92958, 78.05452



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 10.9110477, 78.09713769999999



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37902, 77.89500



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16755, 78.01568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14485, 78.02282



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10084, 78.00555



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.08280, 78.00339



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.05250, 78.05164



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.90793, 78.09916



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.91105, 78.09714



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.84613, 78.12000



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.84321, 78.16436



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.84057, 78.16518



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.83953, 78.18108



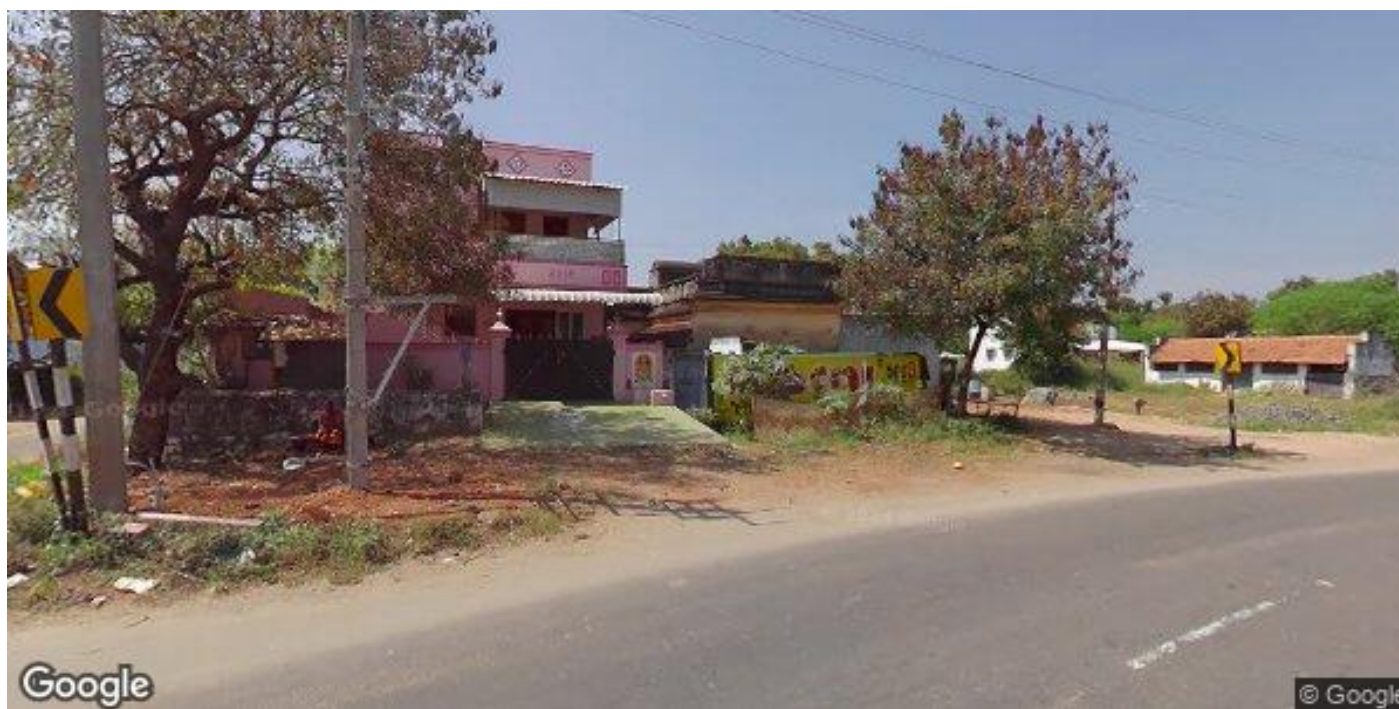
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.83523, 78.18173



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.82892, 78.20166



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.82865, 78.20188



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79274, 78.20214



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79217, 78.20276



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79013, 78.20307



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.78936, 78.20390



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.78767, 78.20438



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.78380, 78.20278



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.77911, 78.21461



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.77530, 78.21486



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.77330, 78.21431



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.77239, 78.21467



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.77215, 78.21560



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.76401, 78.21632



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.75296, 78.22112



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.75178, 78.22246



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.74551, 78.22709



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.73084, 78.23146



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.72910, 78.23136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.72888, 78.23084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.71906, 78.23543



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.71762, 78.23799



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.71635, 78.23810



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.71463, 78.23992



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.71217, 78.23933



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70648, 78.24053

